

PhD Exam. in Analysis Fall 2002

I State the following theorems

1. The Egorov th.
2. The Lebesgue th.
3. The dual of  $L^p(\mu)$ ,  $1 \leq p < \infty$ .
4. The Fubini theorem for  $\mu \times \nu$ -integrable functions
5. The Riesz representation theorem (of linear cont. functional on the space  $C(K)$  of cont. functions on a compact space  $K$ ).

II State and prove one of the following 2 theorems

1. The representation of the dual of  $L^1(\mu)$ .
2. The Fubini theorem for  $\mathcal{I} \times \mathcal{T}$ -measurable functions

III Solve the following problems.

Let  $(X, \Sigma, \mu)$  be a measure space and  $E$  a Banach space

1. Assume  $\mu$  is finite. Let  $\mathcal{F} \subset \Sigma$  be a sub  $\sigma$ -algebra. Prove the existence of the conditional expectation  $E(f|\mathcal{F})$  for functions  $f \in L^1_E(\mu)$ .
2. Let  $\mathcal{R} \subset \Sigma$  be an algebra generating  $\Sigma$  and  $f \in L^1_E(\mu)$ . Prove that there is a sequence  $(f_n)$  of  $\mathcal{R}$ -step functions converging to  $f$ ,  $\mu$ -a.e. and in the mean.
3. Let  $\mathcal{R} \subset \Sigma$  be an algebra ~~and  $f \in L^1_E(\mu)$~~  generating  $\Sigma$  and  $f \in L^1_E(\mu)$ . Prove that if  $\int_A f d\mu = 0$  for every  $A \in \mathcal{R}$ , then  $f = 0$ ,  $\mu$ -a.e.
4. Assume  $\mu$  is finite and let  $f_n \in L^1_E(\mu)$ ,  $n=1,2,\dots$ . Prove that if  $f_n \rightarrow f$  uniformly, then  $f \in L^1_E(\mu)$  and  $f_n \rightarrow f$  in the mean.
5. Let  $(X, \mathcal{F})$ ,  $(Y, \mathcal{T})$  be measurable spaces,  $\mu: \mathcal{F} \rightarrow \mathbb{R}$  and  $\nu: \mathcal{T} \rightarrow \mathbb{R}$   $\sigma$ -additive measures. Prove the existence of the product measure  $\mu \times \nu$  and the equality

$$\int \int f(x,y) d(\mu \times \nu) = \int \int f(x,y) d\mu d\nu$$

Note, Write Complete Computations and explanations.  
Do not use arrows or other unnecessary signs.  
Use words for explanations.

Write complete sentences.

Write complete statements of theorems (including the framework).